培道中學校

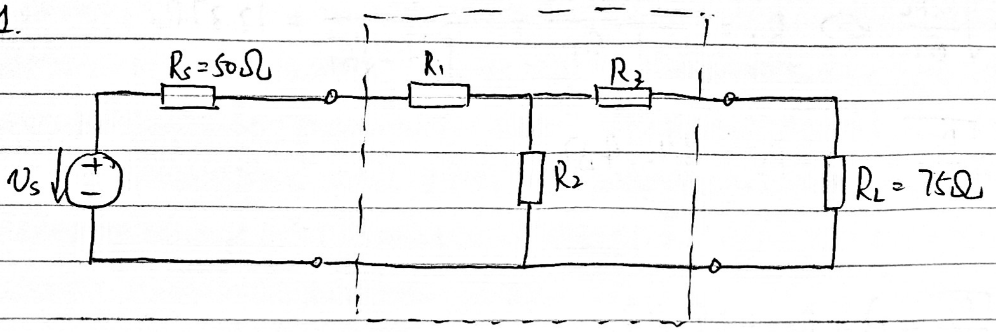
姓名：　　　　　　　　（　　號）

年級：

科目：

成　績

作業 1.



$$Z = \begin{pmatrix} R_1+R_2 & R_2 \\ R_2 & R_2+R_3 \end{pmatrix}, \quad det(Z) = R_1R_2 + R_2R_3 + R_3R_1$$

$$R_s = Z_{01} = \sqrt{\frac{Z_{11}}{Z_{22}} det(Z)} = \sqrt{\frac{R_1+R_2}{R_2+R_3} \cdot (R_1R_2+R_2R_3+R_3R_1)} \quad (1)$$

$$R_L = Z_{02} = \sqrt{\frac{Z_{22}}{Z_{11}} det(Z)} = \sqrt{\frac{R_2+R_3}{R_1+R_2} (R_1R_2+R_2R_3+R_3R_1)} \quad (2)$$

$$G_T = G_{T,max} = \frac{Z_{21}^2}{(\sqrt{|Z|}+\sqrt{Z_{11}Z_{22}})^2} = \frac{R_2^2}{(\sqrt{det(Z)}+\sqrt{(R_1+R_2)(R_2+R_3)})^2} \quad (3)$$

聯立 (1)、(2) 得 $det(Z) = R_sR_L \Rightarrow \dfrac{R_s}{R_L} = \dfrac{R_1+R_2}{R_2+R_3}$ (4)

聯立 (3)、(4) 得 $G_T = \dfrac{R_2^2}{\left(\sqrt{R_sR_L} + (R_2+R_3)\sqrt{\frac{R_s}{R_L}}\right)^2} \Leftrightarrow \sqrt{G_T}\left(\sqrt{R_sR_L} + (R_2+R_3)\sqrt{\frac{R_s}{R_L}}\right) = R_2$

$$R_2\left(\frac{1}{\sqrt{G_T}} - \sqrt{\frac{R_s}{R_L}}\right) = \sqrt{R_sR_L} + R_3\sqrt{\frac{R_s}{R_L}}$$

$$R_3 = R_2\left(\sqrt{\frac{R_L}{G_TR_s}} - 1\right) - R_L \quad (5)$$

聯立 (4)、(5) 得 $\dfrac{R_s}{R_L} = \dfrac{R_1+R_2}{R_2\sqrt{\frac{R_L}{G_TR_s}} - R_L} \Leftrightarrow R_2\sqrt{\dfrac{R_sR_L}{G_T}} - R_sR_L = R_L(R_1+R_2)$

$$R_sR_L = R_1R_2 + (R_1+R_2)R_3 = R_1R_2 + (R_1+R_2)\cdot\left[R_2\left(\sqrt{\frac{R_L}{G_TR_s}} - 1\right) - R_L\right]$$

$$= \left[R_2\left(\sqrt{\frac{R_s}{G_TR_L}} - 1\right) - R_s\right]R_2 + \left[R_2\sqrt{\frac{R_s}{G_TR_L}} - R_s\right]\left[R_2\left(\sqrt{\frac{R_L}{G_TR_s}} - 1\right) - R_L\right]$$

$$= \left(\sqrt{\frac{R_s}{G_TR_L}} + \frac{1}{G_T} - \sqrt{\frac{R_s}{G_TR_L}}\right)R_2^2 - \left(R_s + \sqrt{\frac{R_LR_s}{G_T}} + \sqrt{\frac{R_sR_L}{G_T}} - R_s\right)R_2 + R_sR_L$$

~~所以有~~ $\cancel{\frac{1}{G_T}R_2^2 - 2\sqrt{\frac{R_SR_L}{G_T}}R_2 = 0 \Leftrightarrow R_2 = 2\sqrt{R_SR_LG_T} = 2\sqrt{\frac{R_SR_L}{L}} = 2\times\sqrt{\frac{50\times 75}{100}} = 5\sqrt{6}\,\Omega}$

所以有 $\left(\frac{1}{G_T}-1\right)R_2 = 2\sqrt{\frac{R_SR_L}{G_T}} \Leftrightarrow R_2 = \frac{2\sqrt{R_SR_LG_T}}{1-G_T} = \frac{2\sqrt{R_SR_L}}{\sqrt{\frac{1}{G_T}}-\sqrt{G_T}} = 12.37\Omega$

$$R_3 = \frac{2\sqrt{R_SR_L}}{\sqrt{\frac{1}{G_T}}-\sqrt{G_T}}\left(\sqrt{\frac{R_L}{G_TR_S}}-1\right) - R_L = 64.14\Omega$$

$$R_1 = \frac{2\sqrt{R_SR_L}}{\sqrt{\frac{1}{G_T}}-\sqrt{G_T}}\left(\sqrt{\frac{R_S}{G_TR_L}}-1\right) - R_S = 38.64\Omega$$

因此. $$Z = \begin{pmatrix} \frac{2R_S}{1-G_T}-R_S & \frac{2\sqrt{R_SR_L}}{\sqrt{\frac{1}{G_T}}-\sqrt{G_T}} \\ \frac{2\sqrt{R_SR_L}}{\sqrt{\frac{1}{G_T}}-\sqrt{G_T}} & \frac{2R_L}{1-G_T}-R_L \end{pmatrix} = \begin{pmatrix} 51.01 & 12.37 \\ 12.37 & 76.52 \end{pmatrix}$$

$$Z_{in} = Z_{11} - \frac{Z_{12}Z_{21}}{Z_{22}+R_L} = 50.0\Omega, \quad Z_{out} = Z_{22} - \frac{Z_{12}Z_{21}}{Z_{11}+R_S} = 75.0\Omega$$

$$S_{11} = \frac{Z_{in}-R_S}{Z_{in}+R_S} = 0, \quad S_{22} = \frac{Z_{out}-R_L}{Z_{out}+R_L} = 0$$

$$S_{21} = S_{21} = \sqrt{G_T} = 0.1$$

因此 $$S = \begin{pmatrix} 0 & 0.1 \\ 0.1 & 0 \end{pmatrix}$$

培道中學校

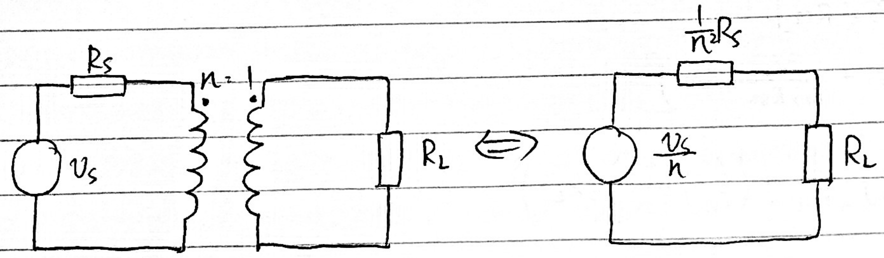
姓名：　　　　　　　　（　號）

年級：

科目：

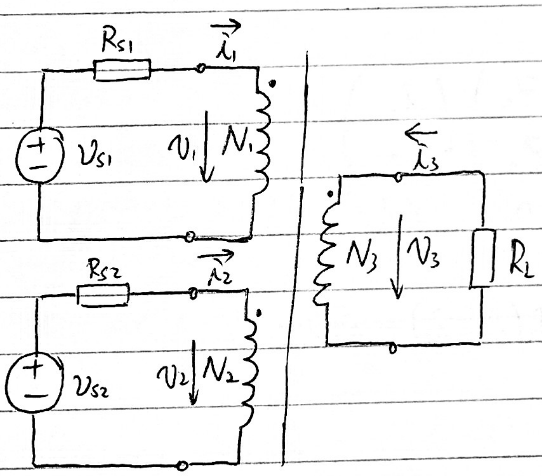
成績

作業 2



當且僅當 $R_L = \frac{1}{n^2}R_S$ 時，負載電阻可獲得最大功率 $= \frac{v_s^2}{4R_S}$（即信源輸入的額定功率）
理想變壓器是無損器件，消耗 0W 功率

作業 3.



$$\begin{pmatrix} v_1 \\ v_2 \\ i_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & \frac{N_1}{N_3} \\ 0 & 0 & \frac{N_2}{N_3} \\ -\frac{N_1}{N_3} & -\frac{N_2}{N_3} & 0 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \\ v_3 \end{pmatrix}$$

$$\begin{aligned} v_3 &= -i_3 R_L \\ &= -\begin{pmatrix} -\frac{N_1}{N_3} & -\frac{N_2}{N_3} \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix} R_L \\ &= \begin{pmatrix} \frac{N_1}{N_3} & \frac{N_2}{N_3} \end{pmatrix} \begin{pmatrix} \frac{v_{S1}-v_1}{R_{S1}} \\ \frac{v_{S2}-v_2}{R_{S2}} \end{pmatrix} R_L \\ &= \begin{pmatrix} \frac{N_1}{N_3} & \frac{N_2}{N_3} \end{pmatrix} \left[\begin{pmatrix} \frac{v_{S1}}{R_{S1}} \\ \frac{v_{S2}}{R_{S2}} \end{pmatrix} - \begin{pmatrix} v_3 \cdot \frac{N_1}{N_3 R_{S1}} \\ v_3 \cdot \frac{N_2}{N_3 R_{S2}} \end{pmatrix} \right] R_L \\ &= \left(\frac{N_1}{N_3 R_{S1}} v_{S1} + \frac{N_2}{N_3 R_{S2}} v_{S2} \right) R_L - \left(\left(\frac{N_1}{N_3}\right)^2 \cdot \frac{1}{R_{S1}} + \left(\frac{N_2}{N_3}\right)^2 \cdot \frac{1}{R_{S2}} \right) v_3 R_L \end{aligned}$$

$$\Leftrightarrow v_3 \left[1 + \left(\frac{N_1}{N_3}\right)^2 \frac{R_L}{R_{S1}} + \left(\frac{N_2}{N_3}\right)^2 \frac{R_L}{R_{S2}} \right] = \frac{N_1 R_L}{N_3 R_{S1}} v_{S1} + \frac{N_2 R_L}{N_3 R_{S2}} v_{S2}$$

$$\Leftrightarrow v_3 = \underbrace{\frac{\frac{N_1}{N_3} \cdot \frac{R_L}{R_{S1}}}{1 + \left(\frac{N_1}{N_3}\right)^2 \frac{R_L}{R_{S1}} + \left(\frac{N_2}{N_3}\right)^2 \frac{R_L}{R_{S2}}}}_{\alpha} v_{S1} + \underbrace{\frac{\frac{N_2}{N_3} \cdot \frac{R_L}{R_{S2}}}{1 + \left(\frac{N_1}{N_3}\right)^2 \frac{R_L}{R_{S1}} + \left(\frac{N_2}{N_3}\right)^2 \frac{R_L}{R_{S2}}}}_{\beta} v_{S2}$$

即 $\frac{\alpha}{\beta} = \frac{N_1}{N_2} \cdot \frac{R_{S2}}{R_{S1}} \propto \frac{N_1}{N_2}$

$$P_L = \frac{v_3^2}{R_L} = \frac{(\alpha v_{s1} + \beta v_{s2})^2}{R_L} = \frac{\alpha^2 v_{s1}^2}{R_L} + \frac{\beta^2 v_{s2}^2}{R_L} + \frac{2\alpha\beta\, v_{s1} v_{s2}}{R_L}$$

$$= \frac{\left(\frac{N_1}{N_3}\right)^2 \cdot \frac{R_L}{R_{s1}^2} v_{s1}^2 + \left(\frac{N_2}{N_3}\right)^2 \cdot \frac{R_L}{R_{s2}^2} v_{s2}^2 + 2\left(\frac{N_1 N_2}{N_3^2}\right) \cdot \frac{R_L}{R_{s1} R_{s2}} v_{s1} v_{s2}}{\left[1 + \left(\frac{N_1}{N_3}\right)^2 \frac{R_L}{R_{s1}} + \left(\frac{N_2}{N_3}\right)^2 \frac{R_L}{R_{s2}}\right]^2}$$

$$= \frac{\left(\frac{N_1}{N_3 R_{s1}} v_{s1} + \frac{N_2}{N_3 R_{s2}} v_{s2}\right)^2}{\left(\frac{1}{\sqrt{R_L}} + \left[\left(\frac{N_1}{N_3}\right)^2 \cdot \frac{1}{R_{s1}} + \left(\frac{N_2}{N_3}\right)^2 \frac{1}{R_{s2}}\right] \sqrt{R_L}\right)^2}$$

當且僅當 $R_L = \dfrac{1}{\left(\frac{N_1}{N_3}\right)^2 \cdot G_{s1} + \left(\frac{N_2}{N_3}\right)^2 R_{s2}}$ 時匹配.

作業4.

(1)
$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 & R & -R \\ -R & 0 & R \\ R & -R & 0 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \\ i_3 \end{pmatrix} = \begin{pmatrix} 0 & Z_0 & -Z_0 \\ -Z_0 & 0 & Z_0 \\ Z_0 & -Z_0 & 0 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \\ i_3 \end{pmatrix}$$

當端口2開路時，$i_2 = 0$
當端口2短路時，$v_2 = 0$
面對上述兩種情況，都有 $P_2 = i_2 v_2 = 0$

加上負阻放大器是無損網絡，所以有 $P_1 + P_3 = 0$
即(1)的結論得證.

(2) 當 $Z_{L2} = -r < 0$ 時，$P_{L3} = P_1 - P_{L2} = P_1 - i_2^2 Z_{L2} > P_1$

$v_2 = -i_2 Z_{L2} = (i_3 - i_1) Z_0 \Leftrightarrow i_2 = \frac{Z_0}{Z_{L2}} i_1 - \frac{Z_0}{Z_{L2}} i_3$

$$\begin{pmatrix} v_1 \\ v_3 \end{pmatrix} = \begin{pmatrix} \frac{Z_0^2}{Z_{L2}} & -\frac{Z_0^2}{Z_{L2}} - Z_0 \\ Z_0 - \frac{Z_0^2}{Z_{L2}} & \frac{Z_0^2}{Z_{L2}} \end{pmatrix} \begin{pmatrix} i_1 \\ i_3 \end{pmatrix}$$

$$Z_{in} = \frac{Z_0^2}{Z_{L2}} - \frac{\left(\frac{Z_0^2}{Z_{L2}}\right)^2 - Z_0^2}{\frac{Z_0^2}{Z_{L2}} + Z_0} = \frac{Z_0^2}{Z_{L2}} - \frac{Z_0^2}{Z_{L2}} + Z_0 = Z_0$$

培道中學校

姓名：　　　　　　　　（　　號）　　　　成績

年級：

科目：

$$Z_{out} = \frac{Z_0^2}{Z_{L2}} - \frac{\left(\frac{Z_0^2}{Z_{L2}} - Z_0\right)\left(\frac{Z_0^2}{Z_{L2}} + Z_0\right)}{\frac{Z_0^2}{Z_{L2}} + Z_0} = Z_0$$

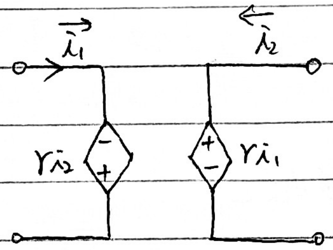
$$\cancel{G_T = \frac{|Z_{21}|^2}{\left(\sqrt{\det(Z)} + \sqrt{Z_{11}Z_{22}}\right)^2} = \frac{\left(Z_0 - \frac{Z_0^2}{Z_{L2}}\right)^2}{\sqrt{\left(\frac{Z_0^2}{Z_{L2}}\right)^2 - \left(\frac{Z_0^2}{Z_{L2}}\right)^2 + Z_0^2 + \frac{Z_0^2}{Z_{L2}}}} =}$$

$$G_T = \frac{\left(Z_0 - \frac{Z_0^2}{Z_{L2}}\right)^2 Z_0^2}{\left(\left(\frac{Z_0^2}{Z_{L2}}\right)^2 - Z_0^2 - \left(\frac{Z_0^2}{Z_{L2}} + Z_0\right)\left(\frac{Z_0^2}{Z_{L2}} + Z_0\right)\right)^2} = \frac{\left(Z_0 - \frac{Z_0^2}{Z_{L2}}\right)^2 Z_0^2}{\left(Z_0 + \frac{Z_0^2}{Z_{L2}}\right)^2 \cdot \left(\frac{2Z_0^2}{Z_{L2}}\right)^2}$$

$$= \frac{\left(1 - \frac{Z_0}{Z_{L2}}\right)^2}{\left(1 + \frac{Z_0}{Z_{L2}}\right)^2 \left(\frac{2Z_0}{Z_{L2}}\right)^2}$$

作業5

(1)



(2) $\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 & -r \\ r & 0 \end{pmatrix}\begin{pmatrix} i_1 \\ i_2 \end{pmatrix} \Rightarrow Z = \begin{pmatrix} 0 & -r \\ r & 0 \end{pmatrix}$

$Y = \begin{pmatrix} 0 & \frac{1}{r} \\ -\frac{1}{r} & 0 \end{pmatrix} \quad (= Z^{-1})$

$\begin{pmatrix} v_1 \\ i_2 \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix}\begin{pmatrix} i_1 \\ v_2 \end{pmatrix} \Leftrightarrow \begin{pmatrix} -r \\ 1 \end{pmatrix} i_2 = \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix}\begin{pmatrix} 1 \\ r \end{pmatrix} i_1$

因為 i_1, i_2 是兩個獨立的項，所以不存在 h 參量

同理可知 不存在 g 參量

$\begin{pmatrix} v_1 \\ i_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}\begin{pmatrix} v_2 \\ -i_2 \end{pmatrix} \Leftrightarrow \begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -i_2 \\ i_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}\begin{pmatrix} r & \\ & 1 \end{pmatrix}\begin{pmatrix} i_1 \\ -i_2 \end{pmatrix}$

$\Leftrightarrow \begin{pmatrix} -i_2 \\ i_1 \end{pmatrix} = \begin{pmatrix} \frac{1}{r} & \\ & 1 \end{pmatrix}\begin{pmatrix} A & B \\ C & D \end{pmatrix}\begin{pmatrix} r & \\ & 1 \end{pmatrix}\begin{pmatrix} i_1 \\ -i_2 \end{pmatrix} = \begin{pmatrix} \frac{A}{r} & \frac{B}{r} \\ C & D \end{pmatrix}\begin{pmatrix} r & \\ & 1 \end{pmatrix}\begin{pmatrix} i_1 \\ -i_2 \end{pmatrix} = \begin{pmatrix} A & \frac{B}{r} \\ rC & D \end{pmatrix}\begin{pmatrix} i_1 \\ -i_2 \end{pmatrix}$

$\Rightarrow -i_2\left(1 - \frac{B}{r}\right) = A i_1$

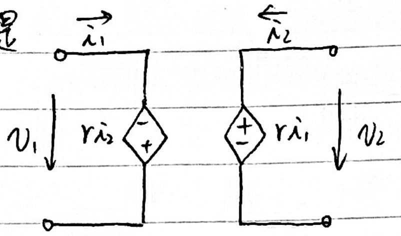
$-i_2 D = (1 - rC) i_1$

因為 i_1, i_2 獨立，所以 $A=0,\ B=r,\ C=\frac{1}{r},\ D=0$

$\Rightarrow \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 0 & r \\ \frac{1}{r} & 0 \end{pmatrix}$

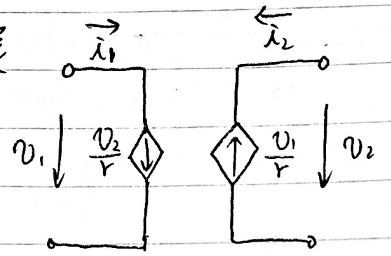
因此 $\begin{pmatrix}a & b\\ c & d\end{pmatrix}=\begin{pmatrix}A & B\\ C & D\end{pmatrix}^{-1}=\begin{pmatrix}0 & r\\ \frac{1}{r} & 0\end{pmatrix}^{-1}=\begin{pmatrix}0 & r\\ \frac{1}{r} & 0\end{pmatrix}$

對 z 參量，其等效電路是



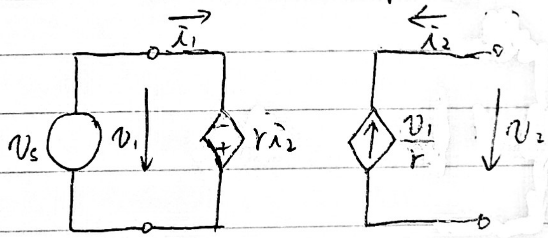
$Z=\begin{pmatrix}0 & -r\\ r & 0\end{pmatrix}$

對 y 參量，其等效電路是



$y=\begin{pmatrix}0 & \frac{1}{r}\\ -\frac{1}{r} & 0\end{pmatrix}$

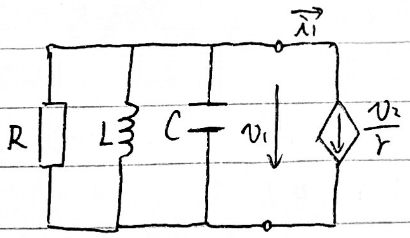
(3) $v_1=0 \iff i_2=-\frac{v_1}{r}=0$ （短路 對應 開路）



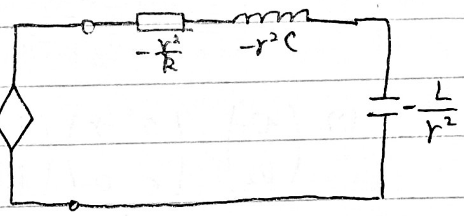
$v_s=v_1 \iff -i_2=\frac{v_1}{r}=\frac{v_s}{r}$

記 $I_s=-i_2=\frac{v_s}{r}$

（恆壓源 對應 恆流源）



$\Longrightarrow$



（並聯RLC 對應 串聯GCL）

$$i_1+\frac{v_1}{R}+C\cdot\frac{dv_1}{dt}+\int_0^t\frac{v_1}{L}dt=0$$

$$\iff v_2=-\frac{r}{R}v_1-rC\cdot\frac{dv_1}{dt}-r\int_0^t\frac{v_1}{L}dt$$

$$\iff v_2=\frac{r^2}{R}i_2+r^2C\frac{di_2}{dt}+\frac{r^2}{L}\int_0^t i_2\,dt$$

(4) 因為 $Z^T=-Z$，所以這是無損器件，也是無源器件

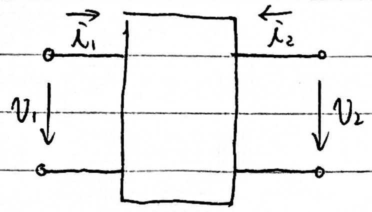
姓名：　　　　　　　　(　　號)

年級：

科目：

成績

作業6



對於 Z 參量，$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix}$

$P_\Sigma = \vec{v}^T \vec{i} = (i_1 \ i_2) \begin{pmatrix} Z_{11} & Z_{21} \\ Z_{12} & Z_{22} \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix} = 0$

取 $\begin{pmatrix} i_1 \\ i_2 \end{pmatrix} = \begin{pmatrix} I \\ 0 \end{pmatrix}$，可得 $P_\Sigma = I^2 Z_{11} = 0 \Leftrightarrow Z_{11} = 0$

取 $\begin{pmatrix} i_1 \\ i_2 \end{pmatrix} = \begin{pmatrix} 0 \\ I \end{pmatrix}$，可得 $P_\Sigma = I^2 Z_{22} = 0 \Leftrightarrow Z_{22} = 0$

此時 $P_\Sigma = (i_1 \ i_2) \begin{pmatrix} i_2 Z_{21} \\ i_1 Z_{12} \end{pmatrix} = i_1 i_2 (Z_{12} + Z_{21}) = 0 \Leftrightarrow Z_{12} = -Z_{21}$

$\begin{pmatrix} i_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix} \begin{pmatrix} v_1 \\ i_2 \end{pmatrix}$，$P_\Sigma = (v_1 \ i_2) \begin{pmatrix} i_1 \\ v_2 \end{pmatrix} = (v_1 \ i_2) \begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix} \begin{pmatrix} v_1 \\ i_2 \end{pmatrix} = 0$

同理可得 $g_{11} = g_{22} = 0$，$g_{12} = -g_{21}$

對於 ABCD 參量，$\begin{pmatrix} v_1 \\ i_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} v_2 \\ -i_2 \end{pmatrix}$，$P_\Sigma = v_1 i_1 + v_2 i_2 = (A \ B) \begin{pmatrix} v_2 \\ -i_2 \end{pmatrix} (C \ D) \begin{pmatrix} v_2 \\ -i_2 \end{pmatrix} + v_2 i_2 \equiv 0$

$P_\Sigma = AC v_2^2 - (AD+BC) v_2 i_2 + BD i_2^2 + v_2 i_2 \equiv 0$

$\Rightarrow AC = BD = 0$，$AD + BC = 1$

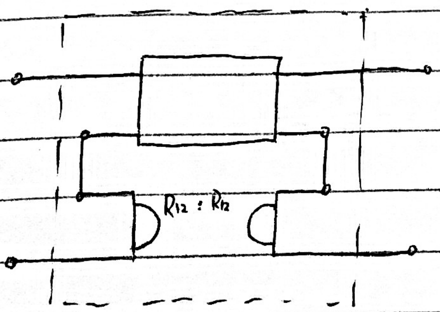
作業7

(1) 網絡有源 $\Leftrightarrow \exists\, i_1, i_2 \in \mathbb{R}$，$0 > P_\Sigma = (i_1 \ i_2) \begin{pmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix} = R_{11} i_1^2 + (R_{12} + R_{21}) i_1 i_2 + R_{22} i_2^2$

$P_\Sigma = R_{11}\left[i_1^2 + \frac{R_{12}+R_{21}}{R_{11}} i_1 i_2\right] + R_{22} i_2^2 = R_{11}\left(i_1 + \frac{R_{12}+R_{21}}{2R_{11}} i_2\right)^2 - \left(R_{11} \cdot \left(\frac{R_{12}+R_{21}}{2R_{11}}\right)^2 - R_{22}\right) i_2^2$

有源性條件是 $\left(\frac{R_{12}+R_{21}}{2R_{11}}\right)^2 > \frac{R_{22}}{R_{11}} \Leftrightarrow (R_{12}+R_{21})^2 > 4R_{11}R_{22}$

(2)

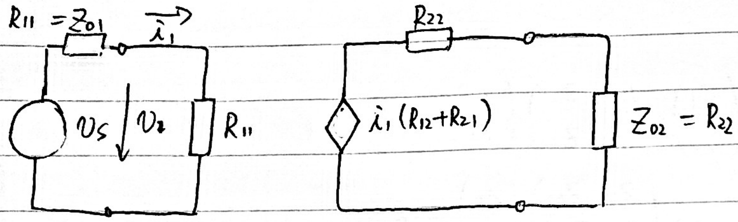


方法：與回旋器串聯

回旋器之參量是 $\begin{pmatrix} 0 & -R_{12} \\ R_{12} & 0 \end{pmatrix}$

(3) 變換後的單向網絡之參量是 $\begin{pmatrix} R_{11} & 0 \\ R_{12}+R_{21} & R_{22} \end{pmatrix}$

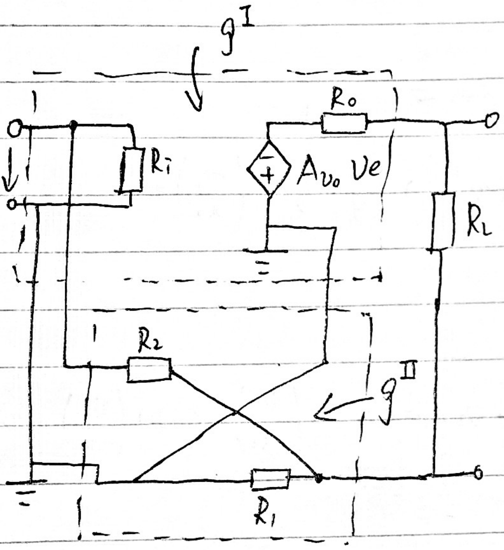
其等效電路是



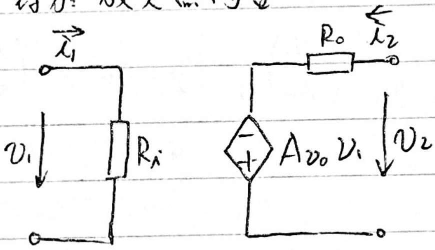
$$G_T=\frac{\dfrac{i_1^2(R_{12}+R_{21})^2}{R_{22}}}{\dfrac{v_s^2}{4R_{11}}}=\frac{\dfrac{i_1^2(R_{12}+R_{21})^2}{4R_{22}}}{\dfrac{i_1^2(2R_{11})^2}{4R_{11}}}=\frac{(R_{12}+R_{21})^2}{4R_{11}R_{22}}>1 \Leftrightarrow (R_{12}+R_{21})^2>4R_{11}R_{22}$$

作業 8

(1)



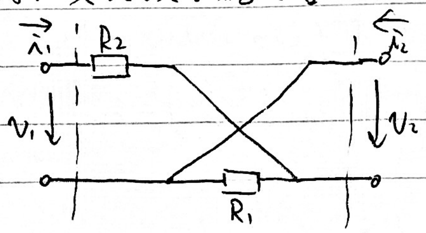
(2) 對於放大器而言



$g_{11}=\left.\dfrac{i_1}{v_1}\right|_{i_2=0}=\dfrac{1}{R_i}$, $g_{21}=\left.\dfrac{v_2}{v_1}\right|_{i_2=0}=-A_{v_o}$

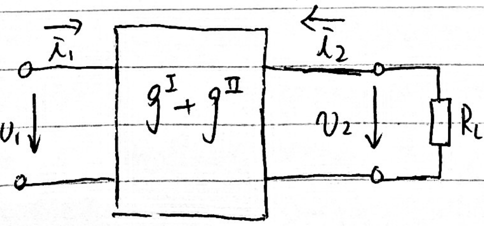
$g_{12}=\left.\dfrac{i_1}{v_2}\right|_{v_1=0}=0$, $g_{22}=\left.\dfrac{v_2}{i_2}\right|_{v_1=0}=R_o$

對於負反饋系統而言



$g_{11}=\left.\dfrac{i_1}{v_1}\right|_{i_2=0}=\dfrac{1}{R_1+R_2}$, $g_{21}=\left.\dfrac{v_2}{v_1}\right|_{i_2=0}=-\dfrac{R_1}{R_1+R_2}$

$g_{12}=\left.\dfrac{i_1}{v_2}\right|_{v_1=0}=\dfrac{1}{R_2}$, $g_{22}=\left.\dfrac{v_2}{i_2}\right|_{v_1=0}=\dfrac{R_1R_2}{R_1+R_2}$



$$g=\begin{pmatrix} \dfrac{1}{R_i}+\dfrac{1}{R_1+R_2} & \dfrac{1}{R_2} \\ -\left(A_{v_o}+\dfrac{R_1}{R_1+R_2}\right) & R_o+\dfrac{R_1R_2}{R_1+R_2} \end{pmatrix}$$